Module 6

Exercise 2(ExpressJs)

1. Create a REST API for User entity with following fields:
   Username
   Password
   FirstName
   LastName
   - Create search bar which will search by user name and will create an autocomplete list of user which fulfil the criteria.

A1.) Code and Output

Module6 > Exercise2 > userRoutes.js > ...

```javascript
const express = require("express");
const router = express.Router();

let users = [];

router.post("/create", (req, res) => {
  try {
    const { userName, password, firstName, lastName } = req.body;

    const user = { userName, password, firstName, lastName };

    users.push(user);

    res.status(201).json(user);

  } catch (err) {
    res.status(500).json({ error: err.message });
  }
});

```

Module6 > Exercise2 > userRoutes.js > ...

```javascript
router.get("/search", (req, res) => {

  const { userName } = req.query;

  if (!userName) {
    return res.status(400).json({ message: "userName query required" });
  }

  const filtered = users.filter(u =>
    u.userName.toLowerCase().includes(userName.toLowerCase())
  );

  res.json(filtered);
});

module.exports = router;
```

Module6 > Exercise2 > server.js > app.listen() callback

```javascript
const express=require('express');
const userRoutes=require('./userRoutes')
const app=express();



app.use(express.json());

app.use('/api/users',userRoutes);

app.listen(8000,()=>{
    console.log("The server listening at 8000");
})
```

http://localhost:8000/api/users/create

Save | No Environment

POST http://localhost:8000/api/users/create Send

Params Authorization Headers (9) Body Scripts Settings Code Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Beautify

```
{
    "userName":"Dhruv145",
    "password":"hello",
    "firstName":"Dhruv",
    "lastName":"Pandey"
}
```

Body Cookies Headers (7) Test Results Status: 201 Created Time: 12 ms Size: 322 B

Pretty Raw Preview JSON

```
{"userName":"Dhruv145","password":"hello","firstName":"Dhruv","lastName":"Pandey"}
```

http://localhost:8000/api/users/search?userName=dh

Save | No Environment

GET http://localhost:8000/api/users/search?userName=dh Send

Params Authorization Headers (7) Body Scripts Settings Code Cookies

Query Params

	Key	Value
☑	userName	dh
	Key	Value

Body Cookies Headers (7) Test Results Status: 200 OK Time: 5 ms Size: 490 B

Pretty Raw Preview JSON

```
[{"userName":"Dhruv145","password":"hello","firstName":"Dhruv","lastName":"Pandey"},
    {"userName":"Dhruv145","password":"hello","firstName":"Dhruv","lastName":"Pandey"},
    {"userName":"Dhipesh145","password":"hello","firstName":"Dhipesh","lastName":"Pandey"}]
```

2. Create a student API which will return list of all students.
    - Create a table which will show all students
    - Create filter for student branch name and on clicking on filters it will show students accordingly.
    - There will be one more column Action in a table which will have a delete option to delete the record.

A2.) Code and Output

Module6 > Exercise2 > studentRoutes.js > ...

```javascript
const express=require('express');

const router=express.Router();

let students = [
  { id: 1, name: "Rahul", branch: "CSE" },
  { id: 2, name: "Aman", branch: "IT" },
  { id: 3, name: "Priya", branch: "CSE" },
  { id: 4, name: "Riya", branch: "ECE" }
];


router.get("/getAll&Filter-students", (req, res) => {

  const { branch } = req.query;

  if (branch) {
    const filtered = students.filter(s => s.branch === branch);
    return res.json(filtered);
  }

  res.json(students);
});
```

Module6 > Exercise2 > studentRoutes.js > ...

```javascript

router.delete("/delete-students/:id", (req, res) => {

  const id = parseInt(req.params.id);

  students = students.filter(s => s.id !== id);

  res.json({ message: "Student deleted" });

});

module.exports=router;
```

All

Name	Branch	Action
Rahul	CSE	Delete
Aman	IT	Delete
Priya	CSE	Delete
Riya	ECE	Delete

CSE

Name	Branch	Action
Rahul	CSE	Delete
Priya	CSE	Delete

All

Name	Branch	Action
Rahul	CSE	Delete
Aman	IT	Delete
Riya	ECE	Delete